Name: _____ Period: ____

Quiz Review: Sections 2.1–2.3

General Physics · Unit 2 · Scalars, vectors, resultants, and components

Show your work: sketch first, write the equation you use, substitute with units, and box your answer. East and north are positive; west and south are negative. Give directions as an angle from a reference direction, such as 37° N of E.

$A_x = A\cos\theta \qquad A_y = A\sin\theta \qquad R = \sqrt{A^2 + B^2} \qquad \theta = \tan^{-1}\left(\frac{\text{opposite}}{\text{adjacent}}\right)$

1. A delivery app shows the information below. Write S (scalar) or V (vector) for each one.

a) "Your package is 4.2 km away." ___ "Drive 4.2 km northwest." ___ "Speed limit: 55 mph." ___

b) "Wind: 20 mph from the west." ___ "Trip time: 12 min." ___ "Push the cart 30 N to the left." ___

2. Mrs. Gordon warms up on a track: 300 m north, then 300 m back south, then 120 m east. The warm-up takes 4.0 min (240 s).

a) Find the distance traveled.

b) Find the change in position (size and direction).

c) Find the average speed and the average velocity.

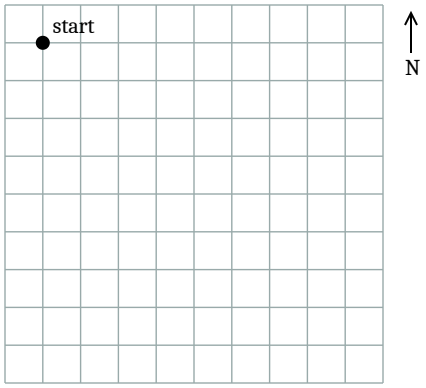
d) Why is the average velocity so much smaller than the average speed?

3. Ivy wanders out of the yard: 70 m east, then 40 m south. On the grid, each square is 10 m and the dot is the yard.

a) Draw both moves tip-to-tail, then draw the resultant. Measure its length and angle.

b) Use the Pythagorean theorem and tangent to calculate the size and direction of the resultant.

c) How far did Ivy travel? Why is that more than your answer to part (b)?



1 square = 10 m

4. Two students pull a stuck wagon with ropes. One pulls with 180 N and the other with 240 N.

a) What is the largest the total pull could be? The smallest? Describe how the ropes point in each case.

b) The ropes pull at right angles: 180 N east and 240 N north. Find the size and direction of the total pull.

5. Mr. Vincenzo shoots a basketball at 9.0 m/s at 50° above the horizontal, toward the hoop on the right.

a) Sketch the arrow and triangle. Then fill in $v_{ix} = 9.0 \cos$ ___ = _____ m/s and $v_{iy} = 9.0 \sin$ ___ = _____ m/s

b) Which component is bigger? If Mr. Vincenzo shoots at 40° instead, which one is bigger then?

6. Dr. Shafer pulls a heavy tire with a rope at 30.0° above the ground. The rope pulls with 500 N.

a) Find the horizontal and vertical components of the pull.

b) A classmate adds them: 433 N + 250 N = 683 N. Explain why that can't be the size of the pull.

7. A wildlife drone's screen shows its velocity components: $v_x = -9.0$ m/s and $v_y = +4.0$ m/s.

a) Sketch the components. Find the drone's speed and the direction it is flying.

b) How long does the drone take to fly 200 m in that direction?

8. **Challenge.** A campsite is 1.50 km from a parking lot at 30.0° north of east. The trail goes due east, then due north to the campsite.

a) How long is each part of the trail?

b) Hikers walk at 1.25 m/s. How many minutes does the trail take? How many minutes would they save if they could walk straight to the campsite at the same speed?